

Advanced Level Mathematics Masterclass

50 Rigorous Questions on Absolute Value Functions, Graphs, and Deductive Solving

Prepared for Advanced Level Mathematics Students

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Abstract

This problem set contains 50 rigorous, examination-style questions tailored for Advanced Level mathematics students. Each question mirrors the structural logic of the reference problem: requiring students to sketch or analyze multiple graphical functions on the same diagram, followed by a deduction ("Hence, solve...") utilizing geometric intersections, inequalities, or algebraic transformations.

Part I: Absolute Value Combinations & Intersections (Q1 – Q15)

1. Sketch the graphs of $y = |x - 1| + 1$ and $y = 5 - |x - 3|$ on the same diagram. Hence, solve the equation $|x - 1| + |x - 3| = 4$.
2. Sketch the graphs of $y = |2x - 3| + 1$ and $y = 6 - |x + 2|$ on the same diagram. Hence, solve the equation $|2x - 3| + |x + 2| = 5$.
3. Sketch the graphs of $y = 3 - |x - 2|$ and $y = |x + 1| - 1$ on the same diagram. Hence, solve the equation $|x - 2| + |x + 1| = 4$.
4. Sketch the graphs of $y = |x - 4| + 2$ and $y = 8 - 2|x - 1|$ on the same diagram. Hence, solve the equation $|x - 4| + 2|x - 1| = 6$.
5. Sketch the graphs of $y = 2|x + 2|$ and $y = 10 - |x - 3|$ on the same diagram. Hence, solve the equation $2|x + 2| + |x - 3| = 10$.
6. Sketch the graphs of $y = |3x - 6| + 2$ and $y = 9 - |x|$ on the same diagram. Hence, solve the equation $|3x - 6| + |x| = 7$.
7. Sketch the graphs of $y = |x - 5| + 3$ and $y = 11 - |x + 1|$ on the same diagram. Hence, solve the equation $|x - 5| + |x + 1| = 8$.
8. Sketch the graphs of $y = 2|x - 1| + 1$ and $y = 7 - |2x - 4|$ on the same diagram. Hence, solve the equation $2|x - 1| + |2x - 4| = 6$.
9. Sketch the graphs of $y = |x + 3| + 2$ and $y = 9 - 2|x - 2|$ on the same diagram. Hence, solve the equation $|x + 3| + 2|x - 2| = 7$.
10. Sketch the graphs of $y = |4 - x| + 1$ and $y = 6 - |x - 2|$ on the same diagram. Hence, solve the equation $|4 - x| + |x - 2| = 5$.
11. Sketch the graphs of $y = 3|x + 1| - 2$ and $y = 7 - |x - 3|$ on the same diagram. Hence, solve the equation $3|x + 1| + |x - 3| = 9$.
12. Sketch the graphs of $y = |2x + 4| + 3$ and $y = 12 - |2x - 2|$ on the same diagram. Hence, solve the equation $|2x + 4| + |2x - 2| = 9$.

13. Sketch the graphs of $y = |x - 3| + |x - 1|$ and $y = 6$ on the same diagram. Hence, solve the inequality $|x - 3| + |x - 1| \leq 6$.
14. Sketch the graphs of $y = |2x + 1| + |x - 2|$ and $y = 7$ on the same diagram. Hence, solve the inequality $|2x + 1| + |x - 2| > 7$.
15. Sketch the graphs of $y = |x + 2| + |x - 4|$ and $y = |2x - 1|$ on the same diagram. Hence, solve the equation $|x + 2| + |x - 4| = |2x - 1|$.

Part II: Quadratic & Modulus Interactions (Q16 – Q30)

16. Sketch the graphs of $y = x^2 - 4x + 3$ and $y = |x - 1|$ on the same diagram. Hence, solve the equation $x^2 - 5x + 4 = |x - 1|$.
17. Sketch the graphs of $y = 9 - x^2$ and $y = |x| + 3$ on the same diagram. Hence, solve the equation $x^2 + |x| - 6 = 0$.
18. Sketch the graphs of $y = |x^2 - 4|$ and $y = 2|x| + 1$ on the same diagram. Hence, solve the equation $|x^2 - 4| - 2|x| - 1 = 0$.
19. Sketch the graphs of $y = x^2 - 2x - 3$ and $y = |2x - 2|$ on the same diagram. Hence, solve the inequality $x^2 - 2x - 3 \leq |2x - 2|$.
20. Sketch the graphs of $y = |x^2 - 2x - 8|$ and $y = 4$ on the same diagram. Hence, solve the equation $|x^2 - 2x - 8| = 4$.
21. Sketch the graphs of $y = 5 - x^2$ and $y = |x - 1| + 1$ on the same diagram. Hence, solve the inequality $x^2 + |x - 1| - 3 \geq 0$.
22. Sketch the graphs of $y = |x^2 - 5x + 6|$ and $y = x + 1$ on the same diagram. Hence, solve the equation $|x^2 - 5x + 6| = x + 1$.
23. Sketch the graphs of $y = x^2 - 6x + 8$ and $y = |x - 3|$ on the same diagram. Hence, solve the equation $|x^2 - 6x + 8| = |x - 3|$.
24. Sketch the graphs of $y = 4 - x^2$ and $y = |x^2 - 4|$ on the same diagram. Hence, solve the equation $4 - x^2 = |x^2 - 4|$.
25. Sketch the graphs of $y = |x^2 - 9|$ and $y = 9 - x^2$ on the same diagram. Hence, deduce the solution set for $|x^2 - 9| + x^2 - 9 = 0$.
26. Sketch the graphs of $y = x^2 + 2x - 3$ and $y = |x^2 + 2x - 3|$ on the same diagram. Hence, solve the equation $x^2 + 2x - 3 = |x^2 + 2x - 3|$.
27. Sketch the graphs of $y = |x^2 - 1| + 2$ and $y = 6 - x^2$ on the same diagram. Hence, solve the inequality $|x^2 - 1| + x^2 \leq 4$.
28. Sketch the graphs of $y = x^2 - 4x$ and $y = |x^2 - 4x|$ on the same diagram. Hence, solve the equation $x^2 - 4x + |x^2 - 4x| = 0$.
29. Sketch the graphs of $y = |x^2 - 3x + 2|$ and $y = 2 - x$ on the same diagram. Hence, solve the equation $|x^2 - 3x + 2| = 2 - x$.
30. Sketch the graphs of $y = x^2 - 5x + 4$ and $y = |x^2 - 5x + 4|$ on the same diagram. Hence, solve the inequality $x^2 - 5x + 4 < |x^2 - 5x + 4|$.

Part III: Rational, Exponential & Trigonometric Logic (Q31 – Q45)

31. Sketch the graphs of $y = \frac{1}{x-1}$ and $y = |x - 2|$ on the same diagram. Hence, state the number of real roots of the equation $(x - 2)|x - 1| = 1$.
32. Sketch the graphs of $y = \left|\frac{2}{x}\right|$ and $y = 3 - |x|$ on the same diagram. Hence, solve the equation $\left|\frac{2}{x}\right| + |x| = 3$.
33. Sketch the graphs of $y = e^x - 1$ and $y = |x - 2|$ on the same diagram. Hence, determine the number of solutions to $e^x - |x - 2| = 1$.
34. Sketch the graphs of $y = \ln x$ and $y = |x - 3|$ on the same diagram. Hence, discuss the number of roots for $\ln x - |x - 3| = 0$.
35. Sketch the graphs of $y = |2^x - 4|$ and $y = 2$ on the same diagram. Hence, solve the equation $|2^x - 4| = 2$.
36. Sketch the graphs of $y = |\sin x|$ and $y = 1 - \frac{|x|}{\pi}$ on the same diagram for $-\pi \leq x \leq \pi$. Hence, find the number of solutions to $|\sin x| + \frac{|x|}{\pi} = 1$.
37. Sketch the graphs of $y = |x^3 - x|$ and $y = 2x$ on the same diagram. Hence, solve the equation $|x^3 - x| = 2x$.
38. Sketch the graphs of $y = |x^3|$ and $y = 4x$ on the same diagram. Hence, solve the equation $|x^3| - 4x = 0$.
39. Sketch the graphs of $y = \sqrt{x}$ and $y = |x - 2|$ on the same diagram. Hence, solve the equation $\sqrt{x} = |x - 2|$.
40. Sketch the graphs of $y = \sqrt{9 - x^2}$ and $y = |x|$ on the same diagram. Hence, solve the equation $\sqrt{9 - x^2} = |x|$.
41. Sketch the graphs of $y = |x^3 - 3x| + 1$ and $y = 3$ on the same diagram. Hence, solve the equation $|x^3 - 3x| = 2$.
42. Sketch the graphs of $y = |x^2 - 2x| - 3$ and $y = 0$ on the same diagram. Hence, solve the equation $|x^2 - 2x| = 3$.
43. Sketch the graphs of $y = \left|\frac{x+1}{x-1}\right|$ and $y = 2$ on the same diagram. Hence, solve the inequality $\left|\frac{x+1}{x-1}\right| \leq 2$.
44. Sketch the graphs of $y = |x^3 - 4x| + 2$ and $y = x^2 + 2$ on the same diagram. Hence, solve the equation $|x^3 - 4x| = x^2$.
45. Sketch the graphs of $y = |\ln(x+1)|$ and $y = 1 - |x|$ on the same diagram. Hence, determine the number of roots for $|\ln(x+1)| + |x| = 1$.

Part IV: Advanced Parametric & Composite Transformations (Q46 – Q50)

46. Let $f(x) = |x - a| + |x - b|$ where $a < b$. Sketch the graph of $f(x)$ for $a = 1, b = 5$ and the line $y = 4$. Hence, deduce the range of values of x for which $|x - 1| + |x - 5| \leq 4$.

47. Sketch the graphs of $y = |x^2 - 2ax|$ for $a = 2$ and $y = x$. Hence, find all values of x satisfying $|x^2 - 4x| = x$.
48. Sketch the graphs of $y = ||x| - 2|$ and $y = 3 - |x|$ on the same diagram. Hence, solve the equation $||x| - 2| + |x| = 3$.
49. Sketch the graphs of $y = |x^2 - 4x + 3|$ and $y = |x - 1| + |x - 3|$ on the same diagram. Hence, solve the equation $|x^2 - 4x + 3| = |x - 1| + |x - 3|$.
50. Given $f(x) = |x - a| + |x - b| + |x - c|$ with $a < b < c$. Sketch the graph of $f(x)$ when $a = 1, b = 3, c = 6$ alongside $y = 5$. Hence, solve the equation $|x - 1| + |x - 3| + |x - 6| = 5$.